A BASE CHANGE OF A DM MORPHISM ALONG A DM STACK OR AN ALGEBRAIC SPACE ADMITS A SURJECTIVE ETALE COVER BY A SCHEME

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The base change permanence of DM-ness. Invoked at the start of the counted proof (source lines 4762-4764). Makes the base-changed morphism DM.
- The statement that a morphism between stacks that is separated over a base and has DM target is DM. Invoked inside the counted proof (source lines 4765-4767). Makes the base-changed stack DM.
- The characterization of DM stacks through the existence of a surjective etale cover by a scheme. Invoked at the end of the counted proof (source lines 4768-4770). Produces the scheme and the surjective etale morphism asserted.

Source reference. Original lemma lemma-base-change-separated: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 703–703. Navigation pointer only: the counted text refers to the Stacks item lemma-base-change-separated, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-separated-over-separated: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 888–888. Navigation pointer only: the counted text refers to the Stacks item lemma-separated-over-separated, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

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Source reference. Original lemma lemma-trivial-implications: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 682–682. Navigation pointer only: the counted text refers to the Stacks item lemma-trivial-implications, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original theorem theorem-DM: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4589–4589. Navigation pointer only: the counted text refers to the Stacks item theorem-DM, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a DM morphism of algebraic stacks. Then*

(1) *For every DM algebraic stack $\mathcal{Z}$ and morphism $\mathcal{Z} \to \mathcal{Y}$ there exists a scheme and a surjective étale morphism $U \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{Z}$.*
(2) *For every algebraic space Z and morphism $Z \to \mathcal{Y}$ there exists a scheme and a surjective étale morphism $U \to \mathcal{X} \times_{\mathcal{Y}} Z$.*

Proof. Proof of (1). As f is DM we see that the base change $\mathcal{X} \times_{\mathcal{Y}} \mathcal{Z} \to \mathcal{Z}$ is DM by Lemma lemma-base-change-separated. Since $\mathcal{Z}$ is DM this implies that $\mathcal{X} \times_{\mathcal{Y}} \mathcal{Z}$ is DM by Lemma lemma-separated-over-separated. Hence there exists a scheme U and a surjective étale morphism $U \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{Z}$, see Theorem theorem-DM. Part (2) is a special case of (1) since an algebraic space (when viewed as an algebraic stack) is DM by Lemma lemma-trivial-implications. □